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Project Title: SQL Mini Project – SmartMart Retail

Introduction

In this project, I worked as a data analyst for SmartMart Retail company.
The company sells products like Electronics, Clothing, and Home Appliances.

The main goal of this project is to analyze:

Best selling products

Sales by cities

Customer behavior

Category performance

Level 1 – Basic SELECT

SELECT * FROM customers;

	customer_id	customer_name	city	age
▶	1	Rahul Sharma	Delhi	28
	2	Priya Verma	Mumbai	32
	3	Amit Singh	Bangalore	25
	4	Neha Kapoor	Delhi	29
	5	Rohit Mehta	Pune	35
	6	Anjali Gupta	Mumbai	27
	7	Karan Patel	Ahmedabad	31
	8	Sneha Reddy	Hyderabad	26
*	NULL	NULL	NULL	NULL

SELECT customer_name, city FROM customers;

Result Grid		Filter Rows:
	customer_name	city
▶	Rahul Sharma	Delhi
	Priya Verma	Mumbai
	Amit Singh	Bangalore
	Neha Kapoor	Delhi
	Rohit Mehta	Pune
	Anjali Gupta	Mumbai
	Karan Patel	Ahmedabad
	Sneha Reddy	Hyderabad

SELECT * FROM products WHERE category='Electronics';

	product_id	product_name	category	price
▶	101	Laptop	Electronics	80000.00
	102	Smartphone	Electronics	40000.00
	103	Headphones	Electronics	2000.00
•	NULL	NULL	NULL	NULL

SELECT * FROM products WHERE price > 10000;

	product_id	product_name	category	price
▶	101	Laptop	Electronics	80000.00
	102	Smartphone	Electronics	40000.00
	107	Refrigerator	Home Appliances	30000.00
	108	Air Conditioner	Home Appliances	45000.00
•	NULL	NULL	NULL	NULL

SELECT DISTINCT category FROM products;

Result Grid			Filter Rows:	Export: 	Wrap C
	category				
▶	Electronics				
	Clothing				
	Home Appliances				

Level 2 – WHERE Conditions

SELECT * FROM customers WHERE city='Delhi';

	customer_id	customer_name	city	age
▶	1	Rahul Sharma	Delhi	28
	4	Neha Kapoor	Delhi	29
•	NULL	NULL	NULL	NULL

SELECT * FROM customers WHERE age > 30;

	customer_id	customer_name	city	age
▶	2	Priya Verma	Mumbai	32
	5	Rohit Mehta	Pune	35
	7	Karan Patel	Ahmedabad	31
•	NULL	NULL	NULL	NULL

SELECT * FROM products WHERE price BETWEEN 2000 AND 50000;

	product_id	product_name	category	price
▶	102	Smartphone	Electronics	40000.00
	103	Headphones	Electronics	2000.00
	105	Jeans	Clothing	2000.00
	106	Microwave	Home Appliances	7000.00
	107	Refrigerator	Home Appliances	30000.00
	108	Air Conditioner	Home Appliances	45000.00
•	NULL	NULL	NULL	NULL

SELECT * FROM sales WHERE order_date > '2024-01-12';

	order_id	customer_id	product_id	quantity	order_date
▶	1005	5	103	4	2024-01-13
	1006	6	106	1	2024-01-14
	1007	7	108	1	2024-01-14
	1008	8	102	1	2024-01-15
	1009	1	103	2	2024-01-16
	1010	2	104	5	2024-01-17
	1011	3	105	1	2024-01-18
	1012	4	101	1	2024-01-19
•	NULL	NULL	NULL	NULL	NULL

SELECT * FROM sales WHERE quantity > 2;

	order_id	customer_id	product_id	quantity	order_date
▶	1003	3	104	3	2024-01-12
	1005	5	103	4	2024-01-13
	1010	2	104	5	2024-01-17
•	NULL	NULL	NULL	NULL	NULL

Level 3 – Sorting & Limiting

SELECT * FROM products ORDER BY price DESC;

	product_id	product_name	category	price
▶	101	Laptop	Electronics	80000.00
	108	Air Conditioner	Home Appliances	45000.00
	102	Smartphone	Electronics	40000.00
	107	Refrigerator	Home Appliances	30000.00
	106	Microwave	Home Appliances	7000.00
	103	Headphones	Electronics	2000.00
	105	Jeans	Clothing	2000.00
	104	T-Shirt	Clothing	800.00
•	NULL	NULL	NULL	NULL

SELECT * FROM products ORDER BY price DESC LIMIT 3;

	product_id	product_name	category	price
▶	101	Laptop	Electronics	80000.00
	108	Air Conditioner	Home Appliances	45000.00
	102	Smartphone	Electronics	40000.00
•	NULL	NULL	NULL	NULL

SELECT * FROM customers ORDER BY age ASC;

	customer_id	customer_name	city	age
▶	3	Amit Singh	Bangalore	25
	8	Sneha Reddy	Hyderabad	26
	6	Anjali Gupta	Mumbai	27
	1	Rahul Sharma	Delhi	28
	4	Neha Kapoor	Delhi	29
	7	Karan Patel	Ahmedabad	31
	2	Priya Verma	Mumbai	32
	5	Rohit Mehta	Pune	35
•	NULL	NULL	NULL	NULL

SELECT * FROM sales ORDER BY order_date DESC LIMIT 5;

	order_id	customer_id	product_id	quantity	order_date
▶	1012	4	101	1	2024-01-19
	1011	3	105	1	2024-01-18
	1010	2	104	5	2024-01-17
	1009	1	103	2	2024-01-16
	1008	8	102	1	2024-01-15
•	NULL	NULL	NULL	NULL	NULL

Level 4 – NULL Handling

UPDATE customers

SET city = NULL

WHERE customer_id = 8;

SELECT * FROM customers WHERE city IS NULL;

Result Grid				
	customer_id	customer_name	city	age
▶	8	Sneha Reddy	NULL	26
•	NULL	NULL	NULL	NULL

SELECT * FROM customers WHERE city IS NOT NULL;

	customer_id	customer_name	city	age
▶	1	Rahul Sharma	Delhi	28
	2	Priya Verma	Mumbai	32
	3	Amit Singh	Bangalore	25
	4	Neha Kapoor	Delhi	29
	5	Rohit Mehta	Pune	35
	6	Anjali Gupta	Mumbai	27
	7	Karan Patel	Ahmedabad	31
•	NULL	NULL	NULL	NULL

Level 5 – Aggregations

SELECT COUNT(*) FROM customers;

	COUNT(*)
▶	8

SELECT AVG(price) FROM products;

Result Grid		 Filter Rows:
	AVG(price)	
	25850.000000	

SELECT MAX(price) FROM products;

	MAX(price)
▶	80000.00

SELECT SUM(quantity) FROM sales;

Result Grid				Filter Row
	SUM(quantity)			
	24			

Level 6 – GROUP BY

SELECT category, COUNT(*) FROM products GROUP BY category;

Result Grid		Filter Rows:
	category	COUNT(*)
▶	Electronics	3
	Clothing	2
	Home Appliances	3

SELECT product_id, SUM(quantity) FROM sales GROUP BY product_id;

Result Grid		Filter Rows:
	product_id	SUM(quantity)
▶	101	2
	102	3
	104	8
	105	3
	103	6
	106	1
	108	1

SELECT category, AVG(price) FROM products GROUP BY category;

Result Grid		Filter Rows:
	category	AVG(price)
▶	Electronics	40666.666667
	Clothing	1400.000000
	Home Appliances	27333.333333

Level 7 – HAVING

```
SELECT category, AVG(price)
FROM products
GROUP BY category
HAVING AVG(price) > 10000;
```

Result Grid		Filter Rows:
	category	AVG(price)
▶	Electronics	40666.666667
	Home Appliances	27333.333333

```
SELECT product_id, SUM(quantity)
FROM sales
GROUP BY product_id
HAVING SUM(quantity) > 3;
```

Result Grid		Filter Rows:
	product_id	SUM(quantity)
▶	104	8
	103	6

```
SELECT category, COUNT(*)
FROM products
GROUP BY category
HAVING COUNT(*) > 2;
```

Result Grid		Filter Rows:
	category	COUNT(*)
▶	Electronics	3
	Home Appliances	3

Business Questions

Most selling category

```
SELECT p.category, SUM(s.quantity) AS total_sales
FROM sales s
JOIN products p ON s.product_id = p.product_id
GROUP BY p.category
ORDER BY total_sales DESC
LIMIT 1;
```

Result Grid			Filter Rows:
	category	total_sales	
▶	Electronics	11	

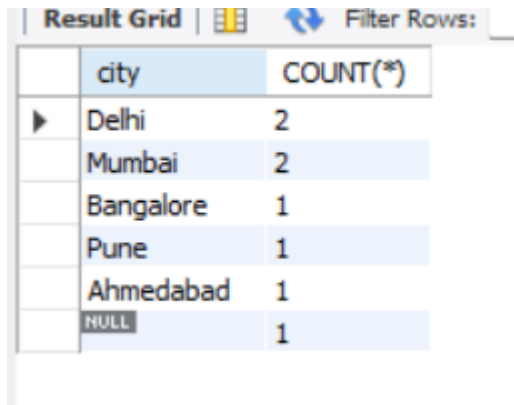
Most sold product

```
SELECT product_id, SUM(quantity) AS total_sold
FROM sales
GROUP BY product_id
ORDER BY total_sold DESC;
```

Result Grid			Filter Rows:
	product_id	total_sold	
▶	104	8	
	103	6	
	102	3	
	105	3	
	101	2	
	106	1	
	108	1	

City with most customers

```
SELECT city, COUNT(*)  
FROM customers  
GROUP BY city  
ORDER BY COUNT(*) DESC;
```

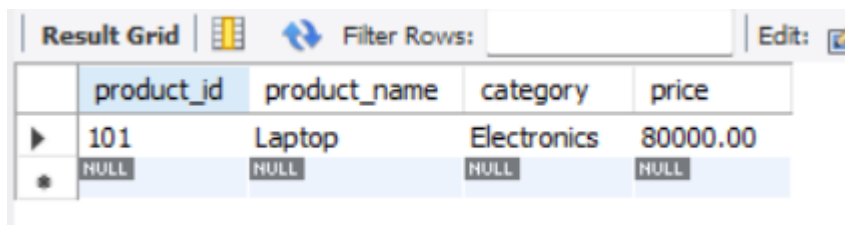


The screenshot shows a database result grid with the following data:

	city	COUNT(*)
▶	Delhi	2
	Mumbai	2
	Bangalore	1
	Pune	1
	Ahmedabad	1
	NULL	1

Highest price product

```
SELECT * FROM products  
ORDER BY price DESC  
LIMIT 1;
```



The screenshot shows a database result grid with the following data:

	product_id	product_name	category	price
▶	101	Laptop	Electronics	80000.00
*	NULL	NULL	NULL	NULL